RESEARCH BRIEF

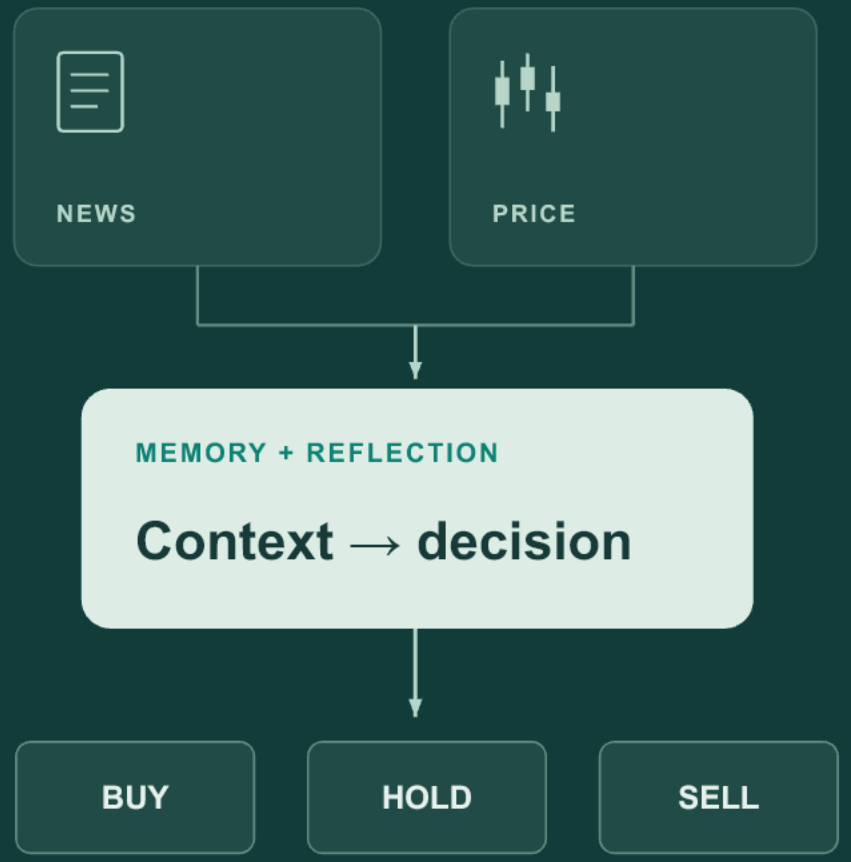
FinAgent

A Multimodal Foundation Agent for Financial Trading

Tool-Augmented, Diversified, and Generalist

> **Beyond forecasting: turn market evidence into action.**

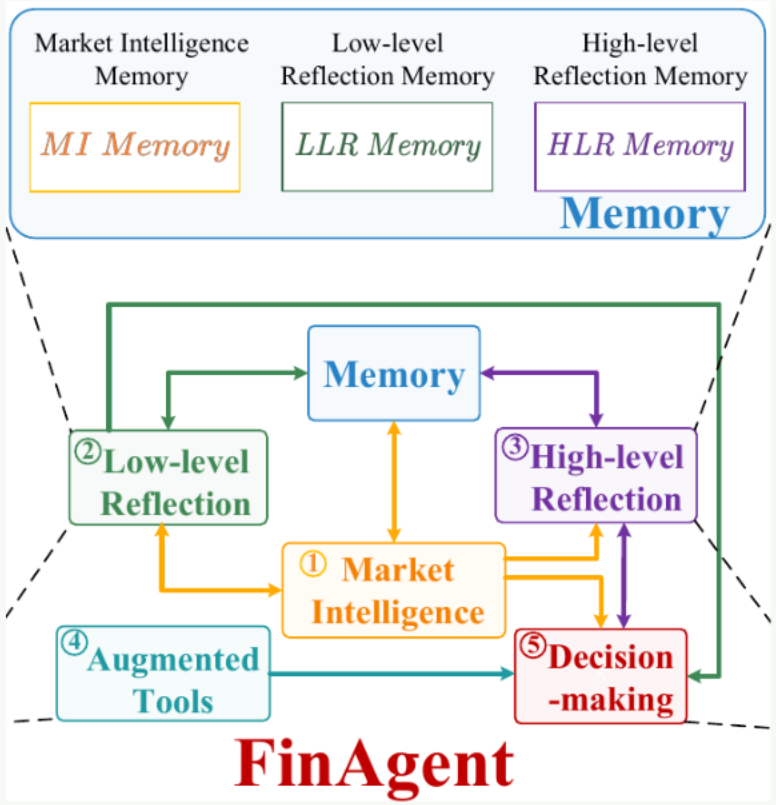
MULTIMODAL IN. REASONED ACTION OUT.



Observe. Recall. Reflect. Act.

FinAgent decision architecture

Core workflow



Market intelligence

Summarize current market information.

Memory and reflection

Retrieve relevant experience and review market signals and previous trades.

Decision with tool support

Combine evidence with technical strategies, expert guidance and trader preferences.

Two reflections. Two different questions.

Separate market interpretation from evaluation of the agent's own behavior.

01 LOW-LEVEL / THE MARKET

Why did prices move?



Market information ↔ price movements

SHORT | MEDIUM | LONG

02 HIGH-LEVEL / THE TRADER

Was the trade sensible?



Past action + rationale + outcome

EXTRACT LESSONS

DIVERSIFIED RETRIEVAL

Decision summary → Trading decision

Retrieval query → Relevant history

Strong on stocks. Not best on every asset.

Annualized return (ARR, %) · Test window: 1 Jun 2023 – 1 Jan 2024



SPOTLIGHT / TSLA

92.27%

annualized return for FinAgent

50.04%
for FinMem

ETH: FinMem is ahead.

5 stocks + Ethereum · 12 baselines · Historical backtest, not a live-trading guarantee

Tool effects differ across assets

Component ablation · Annualized return (ARR, %)

COMPONENT ABLATION

M	L	H	T	TSLA ARR%↑	TSLA SR↑	TSLA MDD%↓	ETHUSD ARR%↑	ETHUSD SR↑	ETHUSD MDD%↓
			√	39.01	0.90	22.54	16.21	0.63	15.93
√				39.27	0.77	30.15	25.97	0.77	24.43
√	√			57.16(+45.56%)	1.02 (+33.14%)	25.77(-14.52%)	52.33(+101.48%)	1.34(+72.99%)	13.59(-44.39%)
√	√	√		89.25(+56.14%)	1.46(+42.86%)	27.62(+7.18%)	54.80(+4.73%)	1.40(+5.09%)	11.74(-13.57%)
√	√	√	√	92.27(+3.38%)	2.01(+37.84%)	12.14(-56.04%)	43.08(-21.39%)	1.18(-16.09%)	12.72(+8.30%)

The last two rows isolate the addition of tools

M: market intelligence L: low-level reflection

H: high-level reflection T: augmented tools

The auxiliary strategies were designed for stocks. Their effect differs on Ethereum.

AFTER ADDING TOOLS

Market fit matters.

TSLA +3.02 pp

ETH −11.72 pp

pp = percentage points of ARR

The contribution is the decision architecture.

Multimodal evidence. Relevant memory. Structured self-review.

01 Combine evidence

Read prices, language and visual context together.

02 Recall what matters

Retrieve relevant experience, not just more text.

03 Review both sides

Interpret the market—and evaluate prior actions.

KEEP THE CLAIM PROPORTIONATE

Promising, not proven everywhere.

Six assets. One historical window.
Further live-market validation is needed.

Supporting data

Backup slide — excluded from the three-minute slide show.

ANNUALIZED RETURN (%)

Asset	FinAgent	Best baseline	Method	Gap (pp)
AAPL	**31.90**	24.84	SAC	+7.06
AMZN	**65.10**	42.93	FinGPT	+22.17
GOOGL	**56.15**	38.29	PPO	+17.86
MSFT	**44.74**	40.58	FinMem	+4.16
TSLA	**92.27**	50.04	FinMem	+42.23
ETH	**43.08**	44.72	FinMem	-1.64

ARR is annualized, not the cumulative test-period return.
Baseline selection uses the highest ARR in each asset column.